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Developmental Reversals in False Memory: Development is Complementary, Not Compensatory

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Abstract

We report the first example of a true complementarity effect in memory development—a situation in which memory for the *same event* simultaneously becomes more and less accurate between early childhood and adulthood. We investigated this paradoxical effect because fuzzy-trace theory predicts that it can occur in paradigms that produce developmental reversals in false memory, which are circumstances in which adults are more likely than children to remember new events as old. The complementarity prediction is: If subjects separately judge whether those same events are new but similar to old ones, adults will be more accurate than children, even though adults are less accurate when they judge whether the items are old. We report four experiments in which children (6- and 10-year-olds), adolescents (14-year-olds), and adults encoded the modal developmental reversal materials, DRM lists. Then, they responded to memory tests on which half the subjects judged whether test items were old and half judged whether the same items were new-similar. The paradoxical complementarity effect was detected in all experiments: The tendency to falsely remember new-similar items as being old increased with development, but so did the tendency to correctly remember them as being new-similar.

Keywords

Developmental reversals; complementarity; compensation; false memory; fuzzy-trace theory

Developmental reversals in false memory are compelling illustrations of the paradoxical nature of human cognition. They are instances in which the baseline tendency to falsely remember events that did not happen is not only high, but it increases during normal development. Also, net accuracy (the ratio of true to false memory) may even decline. These phenomena are puzzling from many points of view. For example, they violate a core principle of cognitive development, the law of positive progression (Brainerd & Reyna, 2012), they run counter to the familiar rule that differentiating reality from fantasy is more difficult for children than for adults (Brainerd, Reyna, & Zember, 2011), and they challenge

policies that govern how the courts interpret the reliability of children's memory reports (Klemfuss & Ceci, 2012).

In this article, we report that developmental reversals are even more counterintuitive than currently supposed because they also violate a principle of logical coherence, which stipulates that memory is compensatory across mutually incompatible reality states. Two reality states are incompatible if an object or event cannot belong to both of them simultaneously. For instance, following the encoding phase of a memory experiment, a test item cannot have been presented *and not presented*; it must be one or the other. The compensation principle merely states that if the tendency to remember that an item belongs to a specific state (say, "presented") increases with development, the tendency to remember that it belongs to some logically incompatible state (say, "not presented") must decrease.

Compensation means that memory development is logically coherent: Increased memory for one reality state is coupled with decreased memory for other incompatible states. The alternative to compensation, developmental *complementarity*, troubles our intuition greatly. Complementarity means that the tendency to remember mutually incompatible states increases with development, so that logical coherence degrades. In the experiments reported below, we shall see that the false memory phenomenon that has dominated developmental reversal research, the Deese/Roediger/McDermott (DRM; Deese, 1959; Roediger & McDermott, 1995) illusion, exhibits precisely this pattern. Before presenting the experiments, we sketch some salient features of prior research that place this pattern in context.

Literature Overview

An intriguing fact about developmental reversals is that they conflict with a century of settled science, which showed that false memories decline during normal development. In addition to some early studies (e.g., Binet, 1900), much evidence of developmental declines has accumulated since the 1990s. To illustrate, developmental declines in spontaneous false memory have been reported using paradigms such as memory for source details (Ackerman, 1992, 1994), for live event sequences (Pipe, Gee, Wilson, & Egerton, 1999; Poole & White, 1991), for metaphorical statements (Reyna & Kiernan, 1995), for mathematics problems (Brainerd & Gordon, 1994), for word lists (Bjorklund & Muir, 1988), and even for nonsense rhymes (Brainerd, Stein, Reyna, 1998). Similarly, declines in implanted false memories have been reported using paradigms as varied as memory for forced confabulations (Ackil & Zaragoza, 1998), for traumatic experiences (Goodman, Quas, Batterman-Faunce, Riddlesberger, & Kuhn, 1994), for crime videos (Bjorklund, Bjorklund, & Brown, 1998), for physical and emotional pain (Bruck, Ceci, Francoeur, E., & Barr, 1995), and for quasi-sexual events (Poole & Lindsay, 1995). Taken together, such findings made the developmental decline rule seem very secure (see various chapters in Bjorklund, 2000).

Appearances were deceiving, however. Brainerd and Reyna (1998) and Ceci and Bruck (1998) were the first to show that fuzzy-trace theory (FTT) predicts that false memories can actually increase with age when they have two properties: (a) Distortion is rooted in comprehending and connecting the semantic content of experience, and (b) it is difficult to

use verbatim memory for actual events to suppress false memories. Brainerd and Mojardin (1998) confirmed that prediction with a paradigm in which false memory was driven by connecting narrative gist across sentences (property a) whose surface forms were so similar that it was difficult to tell exactly which ones had been presented (property b). Soon, several investigators noticed that the paradigm in Table 1, the DRM illusion, has these same two properties, and they found that the illusion intensifies throughout childhood and adolescence (Brainerd, Reyna, Forrest, 2002; Dewhurst & Robinson, 2004; Howe, 2005). In the DRM procedure, subjects encode lists of related words and respond to recognition tests containing three types of probes: old (O) list words, new-similar (NS) words, and new-different (ND) words (cf. Table 1). The task is simply to judge whether each probe is old, with old judgments about O, NS, and ND items being classified as true memories, false memories, and response bias, respectively.

To date, developmental reversals have been detected with an array of paradigms, including memory for categorized photographs (Sloutsky & Fisher, 2004), for historical text (Fazio & Marsh, 2008), for pictures of familiar objects (Ceci, Papierno, & Kulkofsky, 2007), for play sessions (Connolly & Price, 2006), for emotional facial expressions (Fernandez-Dols, Carrera, Barchard, & Gacitua, 2008), for emotional words (Brainerd, Holliday, Reyna, Yang, & Toglia, 2010), for causal inferences (Lyons, Cornoldi, & Ghetti, 2010), for everyday activities (Otgaar, Howe, Brackmann, & Smeets, 2016), and for rumors (Principe, Guiliano, & Root, 2008). However, the DRM illusion has remained the workhorse, with roughly two-thirds of published experiments having relied on it (see Brainerd & Reyna, 2012, Table 1). There are sound practical reasons for this: The paradigm combines high efficiency in false memory production with great simplicity, adaptability, and ease of application with children.

The large archive of DRM research demonstrates that as development unfolds, NS items are increasingly misremembered as being what they are not (old) rather than what they are (new-similar), and by early adulthood, false memory for NS items approaches true memory for O items. It is well established that the magnitude of this trend is influenced by many theoretically-motivated manipulations (for a review, see Brainerd et al., 2011). A few illustrations are encouraging subjects to encode semantic relations (Lampinen, Leding, Reed, & Odegard, 2006; Odegard, Holliday, Brainerd, & Reyna, 2008), increasing the examples of those relations (Sugrue & Hayne, 2006), providing more time to encode list words (Carneiro & Fernandez, 2010), interfering with encoding by dividing attention (Otgaar, Peters, & Howe, 2012), binding list words together with narratives (Dewhurst, Pursglove, & Lewis, 2007), binding list words together with emotional relations (e.g., Howe, 2007), comparing subjects who differ in semantic ability (Weekes, Hamilton, Oakhill, & Holliday, 2007), and comparing subjects who differ in clinical conditions that affect semantic processing (Goodman et al., 2011).

The Present Study

The standard interpretation of developmental reversals in the DRM illusion is that as the brain matures, NS items such as *cold* and *anger* are increasingly perceived as having actually been presented (Brainerd & Reyna, 2012; Paz-Alonso, Ghetti, Donohue, Goodman, & Bunge, 2008). If so, it follows from the compensation principle that the tendency to

correctly perceive that NS items are new-similar must decline with development. However, FTT makes the paradoxical prediction that it can increase (i.e., complementarity). Complementarity effects first became prominent in physics, where they refer to situations in which incompatible empirical states are detected for the same system when each state is measured separately (e.g., Gerlach & Stern, 1922). In the present case, complementarity can result from gist memory, as follows.

FTT assumes that the brain stores dissociated verbatim and gist traces of experience that capture, respectively, events' surface form and semantic content, especially meaning connections among events (e.g., items on the *anger* list are names of emotions). In false memory studies, verbatim traces support true memory for O items *and* suppress false memory for NS items, whereas gist traces support both false memory for NS items and true memory for O items (Abadie, Warnoquer, & Terrier, 2017; Barrouillet, 2015; Bouwmeester & Verkoeijen, 2010; Bruer & Pozzulo, 2014; Lampinen & Odegard, 2006). In other words, verbatim traces are *compensatory* across incompatible reality states (support one, reject the other), but gist traces are *complementary* (support both; Brainerd, Wang, Reyna, & Nakamura, 2015).

In addition to FTT, other explanations of developmental reversals in the DRM illusion were proposed after reversals were reported (see Howe, 2006; Metzger, Warren, Price, Reed, Shelton, & Williams, 2008). However, FTT makes the even more paradoxical complementarity prediction, which does not follow from these later hypotheses. Developmental complementarity falls out of the effects that verbatim and gist memory have when the memory task is changed so that subjects must remember NS items and avoid falsely remembering O and ND items. Now, verbatim traces support true memory for NS items *and* suppress false memory for O items, whereas gist traces support both true memory for NS items and false memory for O items (Brainerd & Reyna, 2018). When performance is dominated by gist, then, whether memory for NS items is true or false depends on whether the old or the new-similar reality state is tested (Brainerd et al., 2015). Therefore, we modified the usual DRM design slightly, so that subjects judged whether NS items (e.g., *cold*) were new-similar, as well as whether they were old. What should development look like?

Remembering NS items to be old on old tests and to be new on new-similar tests is logically contradictory, of course, and hence, our natural intuition is that development must be compensatory—that *cold* will be progressively *less* likely to be remembered as new because we already know from developmental reversal studies that it is progressively *more* likely to be remembered as old. This would mean that although false memories increase with age, at least memory is becoming more logically consistent. According to FTT, however, the noncompensatory nature of gist means that development may not be logically consistent either: Developmental improvements in gist memory make both old and new-similar seem like accurate descriptions of *cold*.

We investigated whether our natural intuition or the theoretical prediction is correct by determining whether, as false memory for NS items increases, true memory for them decreases (compensation) or increases (complementarity). A further theoretical prediction is

that developmental complementarity should be specific to NS items and not extend to O items. Here, research shows that memory for items that are actually encoded is usually dominated by verbatim retrieval (e.g., Abadie et al., 2017; Bouwmeester, & Verkoeijen, 2010), and as we saw, verbatim memory, unlike gist, is compensatory across tests for mutually incompatible reality states.

We report four experiments in which subjects of different ages encoded DRM lists and responded to memory tests on which they made memory judgments about O, NS, and ND items, with half the subjects making traditional old judgments and half making new-similar judgments. Four experiments were included in order to ensure the replicability and generality of what is, after all, a counterintuitive effect. We tracked developmental trends in accuracy for NS items like those in the middle column of Table 1 and for O items like those in the left hand column, between early childhood and adolescence (Experiment 1) and between early childhood and young adulthood (Experiments 2–4).

Experiment 1

Experiment 1 implemented the most commonly used methodology in prior research and covered the age range that has typically been studied. Subjects from three age levels (6-, 10-, and 14-year-olds) were exposed to several DRM lists, followed by recognition tests on which there were three types of items: O, NS, and ND. Relative to most developmental DRM studies, Experiment 1 had two novel features. First and most important, there were two distinct testing conditions. Half of the subjects at each age level participated in a standard O? condition, in which the task was to decide whether each test item was old. The other half participated in an NS? condition, in which the task was to decide whether each test item was new-similar. It should be noted that child NS? tests were developed some years ago by Brainerd and Reyna (1998), and in prior developmental research, children as young as 5-to-7 have performed accurately on such tests (Brainerd & Reyna, 1998; Brainerd, Reyna, & Holliday, 2012; Holliday, Brainerd, & Reyna, 2009). However, our design included manipulation checks to ensure that.

The other novel feature of Experiment 1 is that the memory tests included both types of NS items that are shown in the middle column of Table 1—namely, strong ones (e.g., *cold*, *sleep*, and *anger*), which produce very high levels of DRM false memory in adults, and weak ones (e.g., *frost*, *yawn*, and *mean*), which produce more modest levels. Most DRM studies only include strong NS items, but notice that weak NS items share semantic content with list items. Consequently, FTT predicts the same qualitative trends for both; that is, developmental reversals should occur for weak as well as strong NS items, and developmental complementarity can occur for both.

Method

Subjects.

The subjects in Experiment 1 were 65 first-grade children ($M = 6$ years, 7 months), 64 fifth-grade children ($M = 10$ years, 9 months), and 64 ninth-grade adolescents ($M = 14$ years, 5 month). The choice of sample size was based on a statistical power analysis of prior

developmental DRM experiments that the authors have reported. The analysis showed that sample sizes of 40 per age level would yield a power of $1-\beta = .8$ to detect moderate-sized effects for the between subjects factors of age and condition.

The subjects were recruited from public school that served middle class residential areas in the United Kingdom, and their participation was secured via letters of parental permission. Subjects from each age level were randomly assigned to the O? condition or to the NS? condition, with the restriction that conditions should contain equal numbers of males and females. The research ("Development of False Memory") was classified as exempt by the Institutional Review Board of the University of Kent because it involved simple behavioral testing, which consisted of studying word lists and responding to yes/no memory questions.

Materials.

The words for the study lists and memory tests in all four experiments were drawn from a normed pool of 55 DRM lists, each of which consists of the first 15 forward associates of a missing item (Roediger, Watson, McDermott, & Gallo, 2001). These missing items are the ones that are designated as strong NS in Table 1. The lists are ranked according to the levels of false memory that they produce for missing items (i.e., judging those items to be O). As the study lists for all four experiments, we selected the nine that produce the highest levels of false memory in these norms. (The missing items were *anger*, *cold*, *cup*, *sleep*, *smell*, *soft*, *sweet*, *window*, and *wish*.). Four additional lists were used as sources of items for the instructions and practice tests.

The first 12 words from each of the 9 DRM lists formed the lists that were presented to the subjects at each age level. The lists were presented in 3 cycles of 3 lists per cycle, with a memory test following each cycle of 3 lists. On the memory test that followed each cycle of 3 DRM lists, there were 19 test probes: Three words from each of the 3 lists, the 3 missing words (strong NS items), 1 word that was selected from among the 3 unrepresented words for each list (weak NS items), and 4 words drawn from the 38 DRM lists that were not used as either study lists or as sources of items for the instructions and practice tests and were randomly selected for each subject in each experiment (ND items). The ND items were controls for well-known developmental differences in response bias (e.g., Brainerd & Mojardin, 1998). Thus, over the 3 cycles, the 3 test lists contained a total of 27 O words and 30 new words (9 strong NS, 9 weak NS, and 12 ND).

Procedures.

Quiet rooms were set aside within participating schools, where the research was conducted. Each subject was randomly assigned to either the O? or NS? testing condition upon arrival and was informed that he/she would be exposed to a series of vocabulary lists, after which his/her memory would be tested. It is important to note that the same instructions and testing procedures were administered to all subjects at all age levels in Experiment 1 and also in Experiments 2–4. Hence, none of the developmental trends that we report can be due to age differences in the instructions or procedures that subjects of different ages received because there were none.

Subjects sat at a computer and the experimenter sat nearby. Before any lists were presented, subjects heard the following instructions, via a computer audio recording: “In this study, you will first listen to some lists of vocabulary words. All of them are familiar words that you already know. The lists will be read slowly, one word at a time. Listen to the lists very carefully and pay attention to each word because later on, you will take a memory test. Do you have any questions?” The experimenter then answered any questions. Next, the computer played a recording of 3 12-word DRM lists, being read at a rate of 3 sec per item in a neutral voice with a 10 sec pause between lists. After the first cycle of lists, subjects listened to instructions for the first memory test.

As mentioned, instructions and procedures for matched O? and NS? conditions were developed by Brainerd and Reyna (1998) and have been used with children as young as age 5. For subjects who were assigned to the O? condition, the instructions stressed that their task was to accept only old items that had appeared on the lists of vocabulary words. For subjects who were assigned to the NS? condition, the instructions stressed that their task was to accept only new words that had the same meaning as old words. The instructions for each condition provided subjects with multiple examples of the each of the three types of items that were to be accepted and rejected, which were drawn from short practice lists (four words) that were presented as part of the instructions.

Also as part of the instructions, subjects were required to pass a practice test to demonstrate their understanding and to show that they could apply the instructions on a memory test. Specifically, a short practice list was presented, after which 6 test probes (2 O items, 2 NS items, and 2 ND items) were presented. Subjects received feedback for their responses to each probe. Those who responded correctly to all test probes continued to the main memory test for their condition. Those who made one or more errors received a second practice list and a second practice test. Subjects who failed to perform perfectly on the second practice test were dropped from the experiment. Only three subjects had to be dropped, all of them 6-year-olds.

The main memory test followed successful completion of the practice test. First, subjects listened to an audio recording of instructions that reminded them of how to respond by pressing the Y key on the computer keyboard for items that they wanted to accept or pressing the N key for items that they wanted to reject. Subjects then listened to a computer file of the test items being read in random order. The test was self-paced; the recording did not advance to the next item until subjects had responded Y or N to the current item. There were two more cycles of three DRM lists, followed by reminder test instructions, followed by a self-paced memory test. The practice test was eliminated from the instructional phase of the second and third cycles.

Results

Before reporting the results, we emphasize that it is always necessary to correct raw recognition data such as these for response bias, because as mentioned earlier, levels of yes-saying bias vary developmentally. That was true in the present experiment because false alarm rates for ND items declined steadily with age in both testing conditions, demonstrating higher bias at younger age levels. The three most common methods of

correcting data such as ours for response bias are the two-threshold statistic P_r and the signal detection statistics A' and d' . (Snodgrass & Corwin, 1988). These statistics usually produce the same qualitative results (Brainerd et al., 2015), and consequently, we only report results using the first correction method, P_r . This statistic has the conceptual virtue that corrected scores are probability values rather than dimensionless proportions (A') or Gaussian random variables (d'). However, all of the statistical analyses that we report were also conducted for A' and d' scores, and none of the findings changed.

The mean bias-corrected correct response probabilities for O, strong NS, and weak NS items are reported in Table 2 for the O? and NS? conditions, and the ND false alarm rates that were used in making these corrections may be found in Table 3. As manipulation checks, notice that in the O? condition, the usual developmental reversal pattern is present for strong NS items, and that true memory (accepting O items) also increased between early childhood and adolescence. As another manipulation check, recall that even the youngest children understood the instructions and could execute them on memory tests because they could not participate in the main memory tests unless they performed perfectly on the practice test. Nevertheless, it might be thought that because the NS? instructions, are atypical, the youngest children might have been unable to execute them during the main tests. That hypothesis can be tested by comparing these children's performance in the NS? condition versus the O? condition. Note that if 6-year-olds understood and could implement the instructions, the probability of accepting O items should be higher in the O? condition than in the NS? condition, whereas the opposite should be true for NS items. Both results were obtained. Six-year-olds' acceptance rates for O items were $p(O?) = .60$ and $p(NS?) = .43$, $t(32) = 3.92$, $p < .001$, whereas their acceptance rates for weak NS items were $p(O?) = .04$ and $p(NS?) = .17$, $t(32) = 3.77$, $p < .001$.

The important new findings can be seen in Figure 1A, where *correct response probabilities* in the O? condition (black data points) and the NS? condition (white data points) are plotted for both strong and weak NS items. It can be seen that the developmental trend for these false memory items is not the intuitive compensation pattern, in which NS accuracy would decline with age in both the O? and NS? conditions. Instead, it is the paradoxical complementarity pattern, in which accuracy simultaneously decreases (in the O? condition) and increases (in the NS? condition). In contrast, a different pattern can be seen in Figure 1B for O items. There, the intuitive compensation pattern is present: Memory for list words becomes more accurate with age in both conditions.

In order to test the complementarity pattern in Figure 1A for statistical reliability, we computed a 3 (age) X 2 (condition: O? vs. NS?) X 2 (type of item: strong NS vs. weak NS) analysis of variance (ANOVA) of bias-corrected correct response probabilities for NS items. The first two factors were between-subjects, and the third was within-subjects. If it is true that development is complementary for NS items but compensatory for O items, there will be an Age X Condition cross-over interaction. The specific form of this interaction must be that accuracy decreases with age in the O? condition, while it increases with age in the NS? condition. There was a robust cross-over of just that sort, $F(2, 187) = 178.55$, $MSE = .06$, $\eta_p^2 = .49$. Post hoc tests (Tukey HSD) confirmed what is visually apparent in Figure 1A: In the O? condition, accuracy decreased reliably between the ages of 6 and 10 and between the

ages of 10 and 14, but in the NS? condition, accuracy *increased* reliably between the ages of 6 and 10 and between the ages of 10 and 14. With development, then, subjects became *better* at correctly recognizing that NS items are new-similar, while they simultaneously become *worse* at correctly recognizing that NS items are not old. There was also a Condition X Type of Item interaction, $F(1, 187) = 47.38$, $MSE = .03$, $\eta_p^2 = .19$. Here, post hoc tests revealed something that is also visually apparent in Figure 1A: Accuracy was *higher* for strong than for weak items in the NS? condition, but it was *lower* for strong than for weak items in the O? condition.

Turning to the statistical reliability of the developmental trend for O items (Figure 1B), recall that theory expects that items that were actually presented on lists will exhibit compensation rather than complementarity, for reasons that are concerned with verbatim memory: Memory for list words relies predominantly on verbatim traces, which are compensatory—for O items, they support correct acceptance of O? probes *and* correct rejection of NS? probes (see also Brainerd et al., 2015). It can be seen in Figure 1B that this prediction, too, was confirmed. Statistically, the patterns in Figure 1B mean that an Age X Condition ANOVA of performance on O items should produce a main effect of condition but not an Age X Condition cross-over because accuracy is simply improving with age in both the O? and NS? conditions. There was indeed a main effect for condition, $F(1, 187) = 230.49$, $MSE = .0$, $\eta_p^2 = .55$, $p < .0001$, and there was not an Age X Condition cross-over. There was an Age X Condition interaction of another sort, however, $F(2, 187) = 20.83$, $MSE = .04$, $\eta_p^2 = .22$, $p < .0001$. It was a divergent fan in which the difference between $p(O?)$ and $p(NS?)$ was smaller for 6-year-olds than for 10-year-olds ($\Delta_6 = .17$ and $\Delta_{10} = .44$) and was smaller for 10-year-olds than for 14-year-olds ($\Delta_6 = .44$ and $\Delta_{10} = .68$). In short, with development, subjects became progressively better at remembering that O items are old, *and* that they are not new-similar, which means that the level of logical inconsistency is increasing.

Experiments 2–4

It is well established that the DRM illusion exhibits counterintuitive developmental reversals—that with age, subjects make fewer correct responses to memory probes that ask whether NS items such as *anger* and *cold* are old. Beyond this, the results of Experiment 1 suggest that development is not logically consistent for such items either because, with age, subjects simultaneously make *more* correct responses to memory probes that ask whether these same items are new-similar. In Experiments 2–4, we attempted to replicate this paradoxical pattern, and for the sake of generality, to do so in experiments that varied the design of Experiment 1 in two basic ways.

The most significant variation involved the upper end of the age range that was studied. In Experiment 1, the subjects were children and adolescents. However, the developmental reversal effect has also been studied between adolescence and young adulthood (e.g., Odegard et al., 2008), and this counterintuitive effect continues to grow into the early adult years. It is obviously important to establish whether the same is true of the complementarity—whether the age increase in correct responses when memory probes ask whether NS items are new-similar continues into young adulthood. It should, according to the theoretical

distinctions that predicted the complementarity pattern. That prediction was based on the growth of gist memory, and it is well established that the ability to form meaning connections among items such as the words on DRM lists improves during the adolescent and the early adult years (e.g., Schneider & Pressley, 1997). To evaluate that possibility, we extended the age range of Experiment 1 upward to include young adults in Experiments 2–4.

The other design variation involved the method of presenting DRM lists. We implemented the standard oral method in Experiment 1, which results in the highest levels of false memory for NS items on O? probes (Gallo & Roediger, 2003). Other presentation methods that add visual supports to oral presentations are known to reduce false memory in adults. In particular, three methods of enriching oral presentations with visual supports are known to have this effect—namely, (a) pictures of objects and events named by list words (e.g., Howe, 2006), (b) printed word stems for list words (e.g., Holliday, Brainerd, & Reyna, 2011), and (c) the full printed versions of list words (e.g., Gallo & Roediger, 2003). Two of these visual supports, pictures and word stems, have been investigated in developmental studies, and they are known to suppress false memory in children as well as adults (Holliday et al., 2011; Howe, 2006). They are also known to reduce the magnitude of developmental reversal effects (for a review see, Brainerd & Reyna, 2012). We used each of these methods in turn in Experiments 2–4. More specifically, the subjects once again received oral presentations of DRM lists, but now, they were accompanied by visual depictions (computer screen) of either a picture of the object or action that was named by each word (Experiment 2), a partial printed form of each word (e.g., SN__ for *snow*; Experiment 3), or the full printed form of each word (Experiment 4).

The motivation for this second design variation, like the first, was theoretical. Presentation formatting manipulations are of theoretical interest because they affect the types of memories that are stored. Because these visual support manipulations decrease the tendency to falsely accept O? probes for NS items, the traditional explanation is that they increase verbatim memory, relative to gist memory (Holliday et al., 2011). The fact that they shrink but do not eliminate developmental reversals is explained in the same way (Holliday et al., 2011). This leads to an interesting prediction about subjects' responses to NS items on complementary NS? probes. We saw that the theoretical distinctions that predict developmental reversals also say that verbatim memory can be used to correctly accept such probes (e.g., “*Anger* is definitely new because I clearly remember hearing *mad*, *fear*, and *hate*”). Consequently, these visual support manipulations should increase correct acceptances of NS? probes as they decrease false acceptances of O? probes, especially at older age levels where both acceptance probabilities are high.

Method

Subjects.

A total of 607 subjects participated in these three experiments. The subjects in Experiment 2 were 50 first-grade children ($M = 6$ years, 4 months), 50 fifth-grade children ($M = 10$ year, 11 months), 50 ninth-grade adolescents ($M = 14$ years, 9 months), and 56 undergraduates ($M = 20$ years, 1 month). The subjects in Experiment 3 were 46 first-grade children ($M = 6$ years, 8 months), 44 fifth-grade children ($M = 10$ years, 7 months), 49 ninth-grade

adolescents ($M = 14$ years, 5 months), and 56 undergraduates ($M = 20$ years, 2 months). The subjects in Experiment 4 were 50 first-grade children ($M = 6$ year, 10 months), 50 fifth-grade children ($M = 10$ y, 6 months), 50 ninth-grade adolescents ($M = 14$ years, 11 months), and 56 undergraduates ($M = 20$ 50years, 10 months). The choice of sample size was based on a statistical power analysis of prior developmental DRM experiments that the authors have reported. The analysis showed that sample sizes of 40 per age level would yield a power of $1 - \beta = .8$ to detect moderate-sized effects for the between subjects factors of age and condition.

As in Experiment 1, the subjects at the three youngest age levels in Experiments 2–4 were recruited from public schools that served middleclass residential areas in the United Kingdom, and their participation was secured via letters of parental permission. The subjects at the oldest age level in these experiments were undergraduate volunteers who participated to fulfill a course requirement. As in Experiment 1, the subjects from each age level in Experiments 2–4 were randomly assigned to the O? or NS? condition, with the restriction that each condition should contain equal numbers of males and females. The research was classified as exempt by the Institutional Review Board of the University of Kent because it involved simple behavioral testing, which consisted of studying word lists and responding to yes/no memory questions.

Materials and procedures.

For the subjects at the three youngest age levels in these experiments, as in Experiment 1, the research was conducted in quiet rooms within their schools that had been designated for this purpose. For the adults in each experiment, the research was conducted in university laboratories. The materials and procedures for each experiment were the same as Experiment 1, except for two changes. The first was that during presentation of DRM lists, the oral administration procedure of Experiment 1 was supplemented with visual depictions of the items on a computer screen. As the subjects sat in front of the computer monitor listening to the DRM lists, they viewed visual depictions of the individual items, which were synchronized with the audio recordings. The subjects received the same general instructions as in Experiment 1 at the start of the experiment, but the instructions also informed them as to the nature of the depictions that they would view. In Experiment 2, the subjects were told they would view pictures of objects or actions referred to by each word. These were large pictures that appeared in the center of the screen. In Experiment 3, the subjects were told that they would view a few letters from the printed form each word, such as *__ar*, for *fear*; *__te*, for *hate*, and *__mper* for *temper*. These word stems were printed in lowercase 72-point font and centered on the screen. In Experiment 4, the subjects were told that they would view the full printed form of each word, which was presented in lowercase 72-point font, centered on the screen.

The second change was to reduce the length of the procedure. In Experiment 1, the subjects received three cycles of 3 DRM lists followed by a 25-item self-paced memory test (9 O, 3 strong NS, 3 weak NS, 10 ND), for a total of 75 test items. At different points during the second half of the procedure, some of the youngest children stated that they were becoming fatigued. Although those children's performance did not differ from that of other children,

we decided to shorten the procedure to a single cycle of three DRM lists followed by a single self-paced memory test of 25 items. The 3 lists that were administered to individual subjects in these experiments were randomly selected from among the 12 lists that were used in Experiment 1. As before, subjects received the instructions that were appropriate to their condition, and they were administered a practice test, one on which perfect performance was required in order to proceed to the main memory test. That main memory tests were identical to Experiment 1.

Results

In Table 2, the mean bias-corrected correct response probabilities for O, strong NS, and weak NS items are reported for the O? and NS? conditions of Experiments 2–4. The ND false alarm probabilities that were used to make these bias corrections are reported in Table 3. As manipulation checks, notice that in the O? condition of each experiment, the usual developmental reversal pattern is present for strong NS items (false alarms increase with age), and that true memory displays the usual developmental improvement (hit rates increase with age). As another manipulation check, it will be remembered that even the youngest children in each experiment understood the instructions and could execute them on memory tests because they were required to perform perfectly on practice tests before proceeding. Even the 6-year-olds were able to implement those instructions on the main memory test: A glance at Table 2 reveals that (a) they correctly accepted O items at higher rates in the O? condition than they incorrectly accepted them in the NS? condition, and (b) they incorrectly accepted weak NS items at lower rates in the O? condition than they correctly accepted them in the NS? condition.

The important new findings are displayed and can be seen in Figure 2. There, correct response probabilities in the O? condition and the NS? condition have been plotted for both strong and weak NS items in Experiment 2 (panel A), Experiment 3 (panel B), and Experiment 4 (panel C). It can be seen that as in Experiment 1, our intuition that the development of false memory will be compensatory across mutually incompatible reality states was violated. Instead, the paradoxical complementarity pattern, in which accuracy simultaneously decreases in the O? condition and increases in the NS? condition, was present in all three experiments. That is, with age, subjects become worse at rejecting the possibility that NS items are old but better at accepting the possibility that they are new-similar. In contrast, also as in Experiment 1, it can be seen in Figure 3 the O items exhibited the intuitive compensation pattern: Memory for O items became more accurate with development in both the O? and NS? conditions in all three experiments.

Because the developmental trends were different for NS versus O items, we report the statistical results for true and false memory separately. We begin with the false memory ANOVAs and then move on to parallel true memory ANOVAs.

False memory.

In each experiment, we tested the complementarity pattern in Figure 2 for statistical reliability by computing a 4 (age) X 2 (condition: O? vs. NS?) X 2 (type of item: strong NS vs. weak NS) ANOVA of bias-corrected correct response probabilities for NS items. It will

be recalled that if development is complementary rather than compensatory, this ANOVA must yield an Age X Condition cross-over, such that accuracy decreases with age in the O? condition but increases with age in the NS? condition. There was a robust cross-over of this sort in all three experiments. The detailed results follow.

In Experiment 2, the Age X Condition X Type of Item ANOVA produced a main effect for condition, $F(1, 197) = 71.75$, $MSE = .09$, $\eta_p^2 = .27$, $p < .0001$, and it also produced the crucial Age X Condition interaction, $F(3, 197) = 60.76$, $MSE = .09$, $\eta_p^2 = .44$, $p < .0001$. Post hoc analysis (Tukey HSD) of the interaction confirmed the complementarity pattern that is suggested by the curves in Figure 2A. Specifically, accuracy decreased reliably between age 6 and young adulthood in the O? condition, but it increased reliably in the NS? condition. There was also a small Age X Condition X Type of Item interaction, $F(3, 197) = 4.05$, $MSE = .06$, $\eta_p^2 = .06$, $p < .008$. Post hoc analysis of that interaction revealed that the complementarity pattern was more pronounced for strong NS items than for weak NS items.

Turning to Experiment 3, the Age X Condition X Type of Item ANOVA produced a main effect for condition, $F(1, 189) = 115.54$, $MSE = .10$, $\eta_p^2 = .38$, $p < .0001$, along with the crucial Age X Condition interaction, $F(3, 189) = 36.52$, $MSE = .09$, $\eta_p^2 = .37$, $p < .0001$. Post hoc analysis of the interaction revealed what is obvious in Figure 2B—namely, that accuracy decreased reliably between age 6 and young adulthood in the O? condition, but it increased reliably in the NS? condition. The Age X Condition X Type of Item interaction that was detected in Experiment 2 was not present in this experiment.

In Experiment 4, the Age X Condition X Type of Item ANOVA produced a main effect for condition, $F(1, 196) = 55.87$, $MSE = .10$, $\eta_p^2 = .22$, $p < .0001$, plus the crucial Age X Condition interaction, $F(3, 189) = 35.19$, $MSE = .10$, $\eta_p^2 = .35$, $p < .0001$. As in all prior experiments, post hoc analysis of the interaction revealed that accuracy decreased reliably between age 6 and young adulthood in the O? condition, but it increased reliably in the NS? condition. The small Age X Condition X Type of Item interaction in Experiment 2 was not detected in this experiment.

Finally, we mentioned that the three visual support manipulations should increase correct acceptances of NS? probes as they decrease false acceptances of O? probes, especially at older age level where false acceptance rates are high. We examined this prediction for the 14-year-olds in Experiment 1 (the oldest age level in that experiment) and the 14-year-olds in these experiments. The data were consistent with it. The false acceptance probability for O? probes dropped from .75 in Experiment 1 to an average of .52 in Experiments 2–4, while the correct acceptance probability for NS? probes increased from .43 to an average of .55. Both changes were reliable.

Summing up, complementarity was the core finding about the development of false memory. In each experiment, for both strong and weak NS items, subjects became progressively *better* with age at correctly recognizing that NS items are new-similar, but they become progressively *worse* at correctly recognizing that NS items are not old. In addition, there was always a condition main effect. As can be seen in Figure 2, this was due to the fact that at all

ages, performance was more accurate on average in the NS? condition than in the O? condition.

True memory.

Next, we consider the statistical reliability of the developmental trends for O items that appear in Figure 3. We know that theory predicts that for items that were actually presented on DRM lists, subjects will exhibit compensation rather than complementarity. That was the pattern that was observed in all three experiments.

Inspection of the curves in Figure 3 suggests that (a) accuracy improved with age in both conditions, and (b) performance was more accurate on average in the O? condition than in the NS? condition. Consistent with a, 3(age) X 2 (condition) ANOVAs of the bias-corrected correct response probabilities produced a main effect for age in Experiment 2, $F(3, 197) = 12.00$, $MSE = .08$, $\eta_p^2 = .15$, $p < .0001$, in Experiment 3, $F(3, 189) = 14.05$, $MSE = .05$, $\eta_p^2 = .18$, $p < .0001$, and in Experiment 4, $F(3, 196) = 8.61$, $MSE = .07$, $\eta_p^2 = .12$, $p < .0001$. Further, unlike the false memory ANOVAs, the Age X Condition interaction was not reliable in any of these experiments, which confirms compensation as the core developmental trend for true memory. Finally, there was also a main effect for condition in Experiment 2, $F(1, 197) = 23.04$, $MSE = .08$, $\eta_p^2 = .11$, $p < .0001$, in Experiment 3, $F(1, 197) = 21.74$, $MSE = .08$, $\eta_p^2 = .18$, $p < .0001$, and in Experiment 4, $F(1, 196) = 29.83$, $MSE = .07$, $\eta_p^2 = .13$, $p < .0001$.

In short, the developmental trend across all three experiments was the intuitive one for items that had actually been presented. Between early childhood and young adulthood, subjects became progressively better at recognizing that old items were old and that old items were not new-similar. This developmental growth in accuracy was comparable for the two types of remembering because the Age X Condition interaction was not reliable in any of the experiments.

General Discussion

In developmental studies of false memory, principal interest attaches to how accurately subjects of different ages perform on a single type of memory test for a focal class of items. The memory test asks subjects to remember old (O) items and avoid misremembering new items. The focal items are new-similar (NS) ones that preserve some of the salient content of O items, usually their meaning or some contextual details associated with their presentation. In our research, we shifted attention away from this conventional measure of false memory for NS items toward the developmental *relation* between false and true memory on tests for the logically incompatible old and new-similar reality states. We explored the possibility that this relation violates logical consistency, by exhibiting developmental complementarity rather than developmental compensation. Complementarity was observed with the types of NS items that are administered in the DRM illusion: The tendency to *correctly* remember them as being new-similar increased as the tendency to incorrectly remember them as old increased.

It is easier to appreciate just how contradictory this is by considering a concrete example that involves the three strong NS items in Table 1. On the one hand, suppose that we only have access to data from traditional old tests. If so, we would say those data show that *anger*, *cold*, and *sleep* are increasingly perceived as having been presented on DRM lists because they are increasingly accepted as old. This would be the obvious conclusion if only the lower curves in Figures 1A, 2A, 2B, and 2C were available. On the other hand, suppose that we only have access to data from new-similar tests. Now, we would say that *anger*, *cold*, and *sleep* are increasingly perceived as having *not* been presented on DRM lists because they are increasingly accepted as being new-similar. This would be the obvious conclusion if only the upper curves in Figures 1A, 2A, 2B, and 2C were available.

Because developmental complementarity is a paradoxical result, we sought to generate substantial evidence on whether this may pattern hold for the DRM illusion. Taken as a whole, our findings provide an existence proof of complementarity for this modal developmental reversal paradigm. In Experiment 1, we used the most common methodology for studying the DRM illusion, which is to present lists orally before administering memory tests. We found that between early childhood and mid-adolescence, subjects became progressively better at recognizing that strong and weak NS items are new-similar, but they simultaneously became worse at recognizing that the same items are not old. In Experiments 2–4, we broadened the age range to include young adults, as well as 6-, 10-, and 14-year-olds. We also varied the DRM methodology somewhat by adding three types of visual supports to oral presentations of lists (pictures, word stems, and full printed words), thereby providing a test of the generality of the complementarity pattern in Experiment 1. Visual supports did, indeed, weaken the DRM illusion. For example, among 14-year-olds in Experiment 1, the false acceptance rate on old tests for strong NS items was .74, whereas the corresponding acceptance rates for 14-year-olds in the last three experiments were .51, .46, and .48, respectively. Nevertheless, developmental complementarity was present in all three experiments, for both strong and weak NS items. A difference between the experiments is that complementarity was more marked for strong than for weak NS items in Experiments 1 and 2 but not in Experiments 3 and 4. This may be due to differences among these experiments in presentation methods. It also may be due to range restriction, as overall levels of complementarity were lower in the last two experiments.

In addition to the amount of evidence, another factor that bears on our level of confidence was in the complementarity pattern is that it was not an accidental finding. Rather, this possibility was revealed by the same theoretical machinery that first predicted developmental reversals—namely, FTT's distinctions between dissociated verbatim and gist traces of experience. Both types of representations are interpreted as signs of prior presentation on conventional old tests, but verbatim traces can suppress that tendency for new-similar items. FTT treats developmental reversals as situations in which improvements in gist memory increase the tendency to accept NS items as being old, and parallel improvements in verbatim memory are ineffective in suppressing this tendency (Brainerd et al., 2011). Another feature of gist, the one that points to complementarity, is that it is a noncompensatory form of memory that is congruent with incompatible reality states (Brainerd et al., 2015), so that developmental improvements in gist can lead to increased memory for mutually incompatible states.

Beyond developmental complementarity for NS items, there is another striking complementarity prediction, which is about developmental trends in overall memory accuracy for the *combined data* of O and NS items. Based on the results of Experiments 1–4, overall accuracy for the combined data (a) should increase with age when subjects judge whether both types of items are new-similar but (b) should decrease with age when they judge whether these same items are old. For the two testing conditions, we define overall accuracy for O and NS items in the obvious way as $p_{NS}(NS?) \div [p_O(NS?) + p_{NS}(NS?)]$ when subjects judge whether these items are new-similar and as $p_O(O?) \div [p_O(O?) + p_{NS}(O?)]$ when subjects judge whether these items are old. $p_O(NS?)$ and $p_{NS}(NS?)$ are the probabilities of judging that O and NS items, respectively, are new-similar, whereas $p_O(O?)$ and $p_{NS}(O?)$ are the probabilities of judging that O and NS items, respectively, are old. Thus, for each testing condition, the overall accuracy measure is simply the hit rate divided by the hit rate plus the false alarm rate.

We already know that normal memory development can be complementary for NS items and compensatory for O items, and from that fact, overall accuracy for both types of items should decline when subjects judge whether these items are old but should improve when subjects judge whether these same items are new-similar. The results, pooled across the four experiments, are plotted in Figure 4. As expected, overall accuracy declines between early childhood and young adulthood when subjects judge whether both types of items are old, but overall accuracy improves during the same age range when subjects judge whether both types of items are new-similar. The key implication for understanding memory development is that whether the accuracy of children's memories increases with age is not absolute but, rather, depends upon the reality state that they are asked to remember.

Next, we offer two suggestions about the direction of future research on developmental complementarity. The first is about the DRM illusion, and the second is about other developmental reversal paradigms. With respect to the DRM illusion, we focused on a simple empirical objective in our experiments, which was to generate a substantial amount of data that spanned the early childhood to young adulthood age range. That seemed to be the appropriate first step, considering that complementarity is such a counterintuitive possibility. This focus meant that there was no attempt to test theoretical explanations of complementarity via manipulations that targeted specific processes, such as children's ability to extract the semantic content of DRM lists or preserve verbatim traces of list words. In view of the support for complementarity that emerged, attention can now turn to testing theoretical explanations. Because complementarity was predicted by the process distinctions that originally predicted developmental reversals, an obvious approach would be (a) to determine whether those particular processes are actually tied to variations in complementarity and perhaps (b) to determine whether the processes posited in other accounts of developmental reversals in the DRM illusion (e.g., Howe, 2006; Metzger et al., 2008) are tied to variations in complementarity. Concerning b, we mentioned earlier that unlike FTT, these other accounts do not predict the complementarity phenomenon. However, it may be that proponents of those accounts will be able to elaborate their assumptions in order to explain complementarity after the fact. If that proves to be possible, it would be sensible to investigate whether complementarity varies as a function of the processes that are posited in those accounts.

Turning to the second suggestion, we investigated complementarity in the context of the DRM illusion because it is by far the most widely studied example of developmental reversals. As noted earlier, however, there are several other examples that involve quite different procedures, including ones that focus on remembering emotional events. A key reason that such varied paradigms have been studied is that developmental reversals are counterintuitive, and hence, it is essential to determine whether they are confined to the DRM paradigm or whether they can be detected with other procedures that satisfy the two theoretical conditions that first predicted them. Obviously, the same point applies to developmental complementarity. Similar to the early years of research on developmental reversals, current evidence of developmental complementarity is confined to the DRM paradigm. Theoretically, complementarity may also occur with other paradigms that display developmental reversals. Whether it does is an open empirical question.

Finally, a measurement question should be briefly considered. In all of our experiments, the complementarity phenomenon was measured with a between-subjects procedure in which half the subjects judged whether test items were old and half judged whether the same items were new-similar. Readers may be wondering whether complementarity might not be observed with a comparable within-subjects procedure, in which each subject makes old judgments about half the items and new-similar judgments about the other half. Insofar as children and adolescents are concerned, this remains an open question. However, the answer for adults is known. Recently, we reported a series of experiments in which complementarity was studied with both the between- and within-subject procedures (Brainerd & Reyna, 2018). The basic phenomenon was detected with both procedures, and indeed, the observed levels of complementarity were similar.

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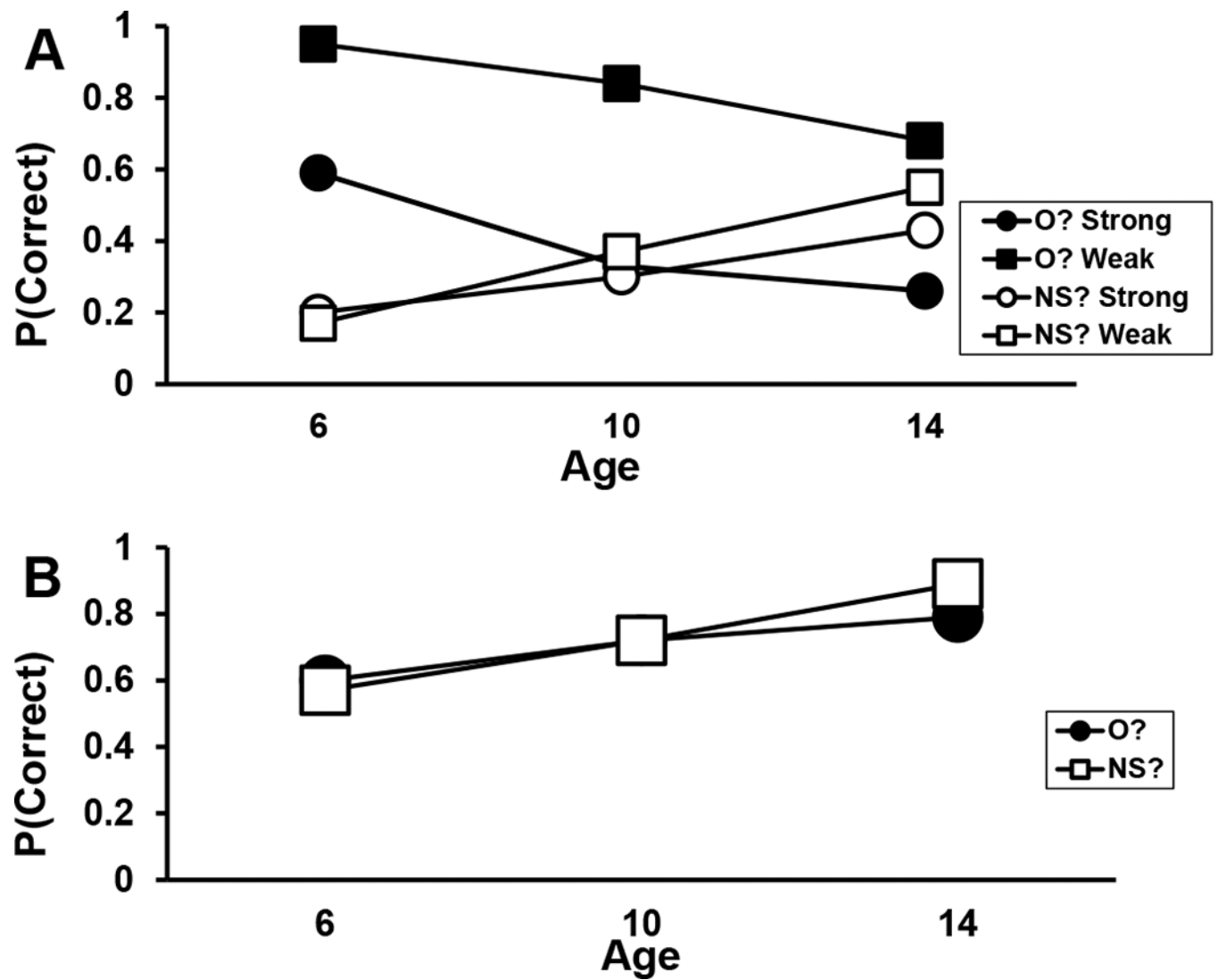


Figure 1. Developmental complementarity in memory accuracy for new-similar (NS) items (Panel A) and developmental compensation in memory accuracy for old (O) items (Panel B) in the DRM illusion in Experiment 1. O? = the old recognition condition, and NS? = the new-similar recognition condition.

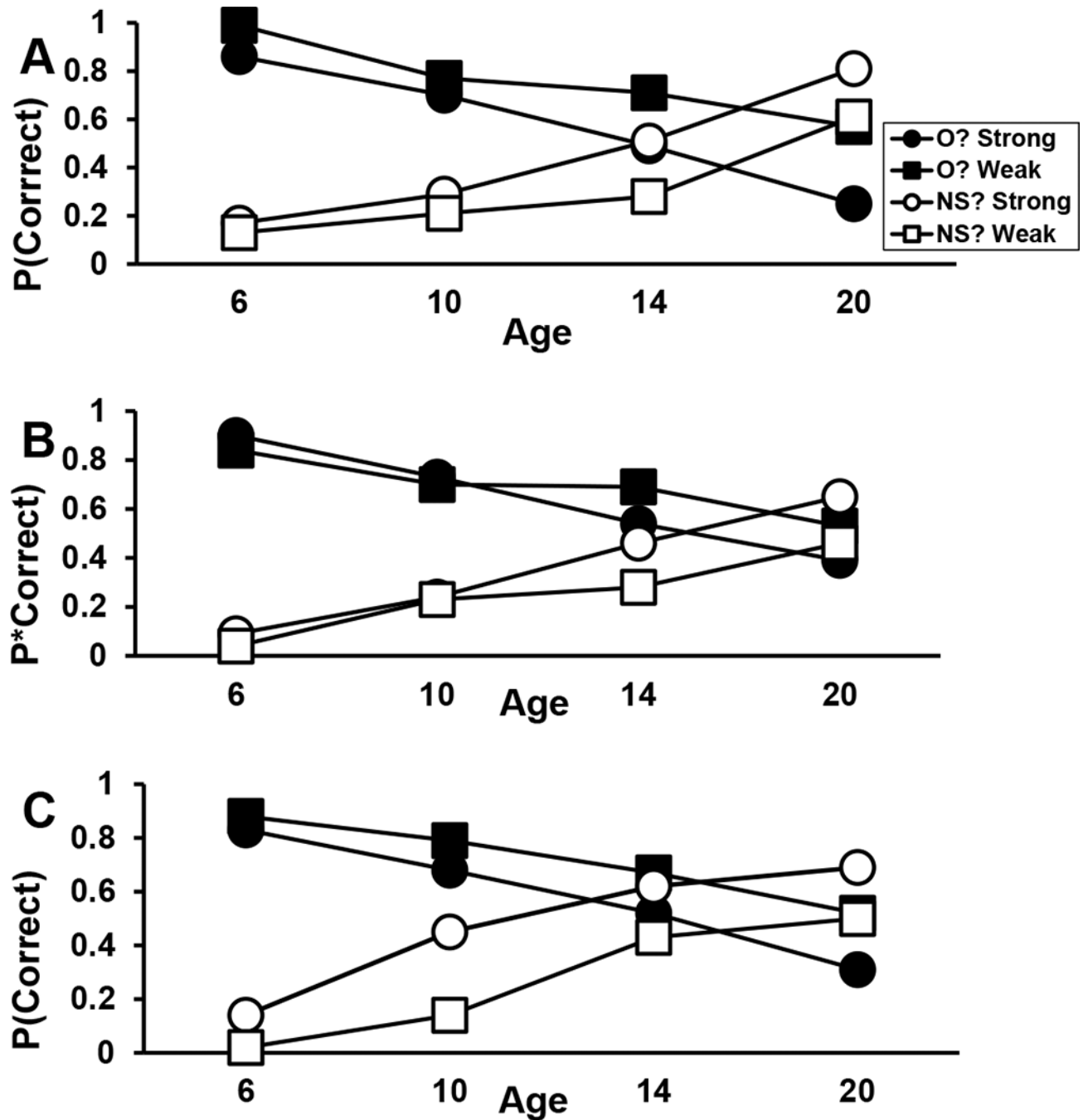


Figure 2. Developmental complementarity in memory accuracy for NS items in the DRM illusion. Accuracy declines with age when subjects judge whether these items are old (O?), but it improves with age when they judge whether these items are NS (NS?). Experiment 2 = Panel A, Experiment 3 = Panel B, and Experiment 4 = Panel C.

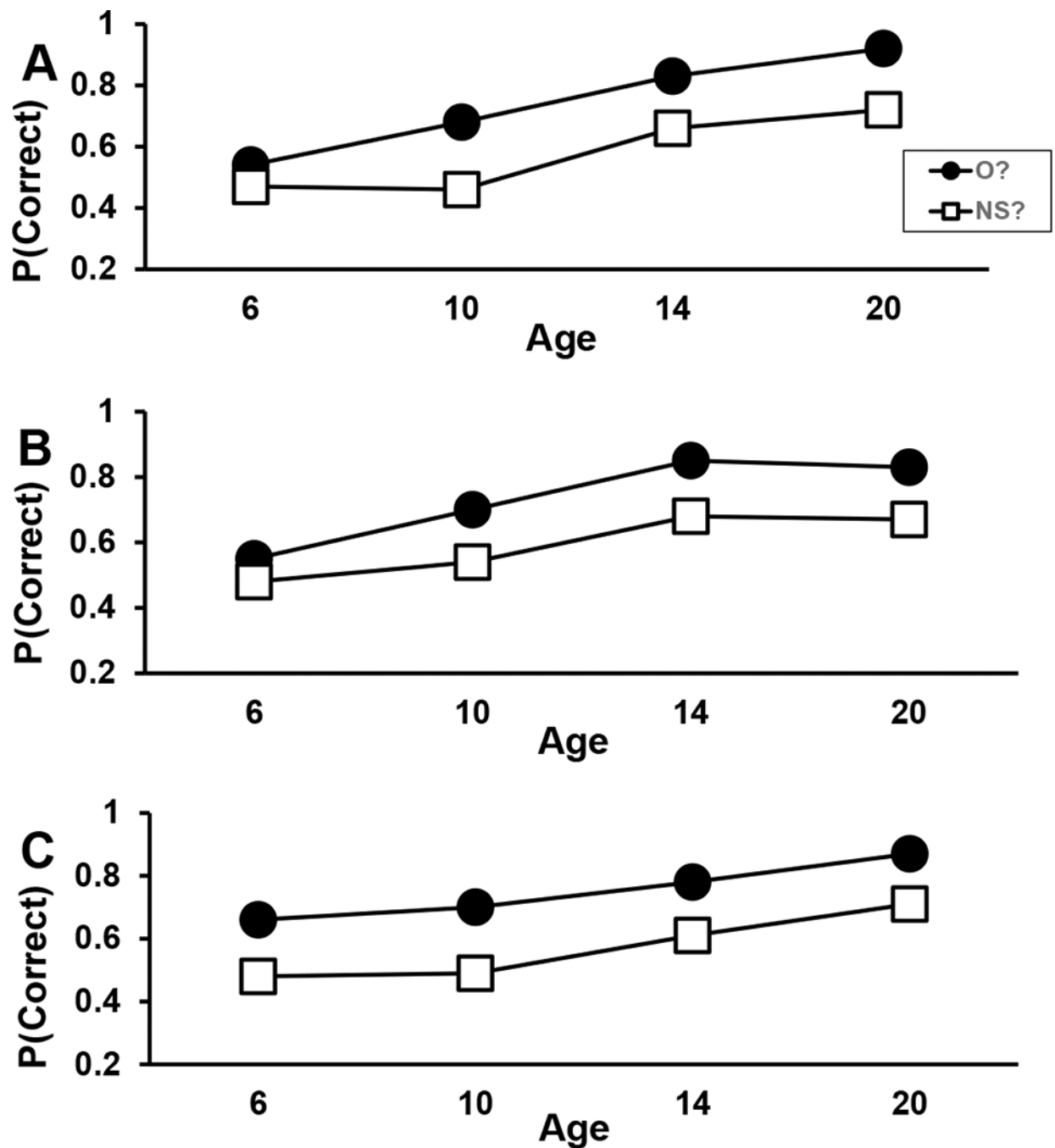


Figure 3.

Developmental compensation in memory accuracy for old (O) items in the DRM illusion. Accuracy increases with age regardless of whether subjects judge whether O items are old (O?) or new-similar (NS?). Experiment 2 = Panel A, Experiment 3 = Panel B, and Experiment 4 = Panel C.

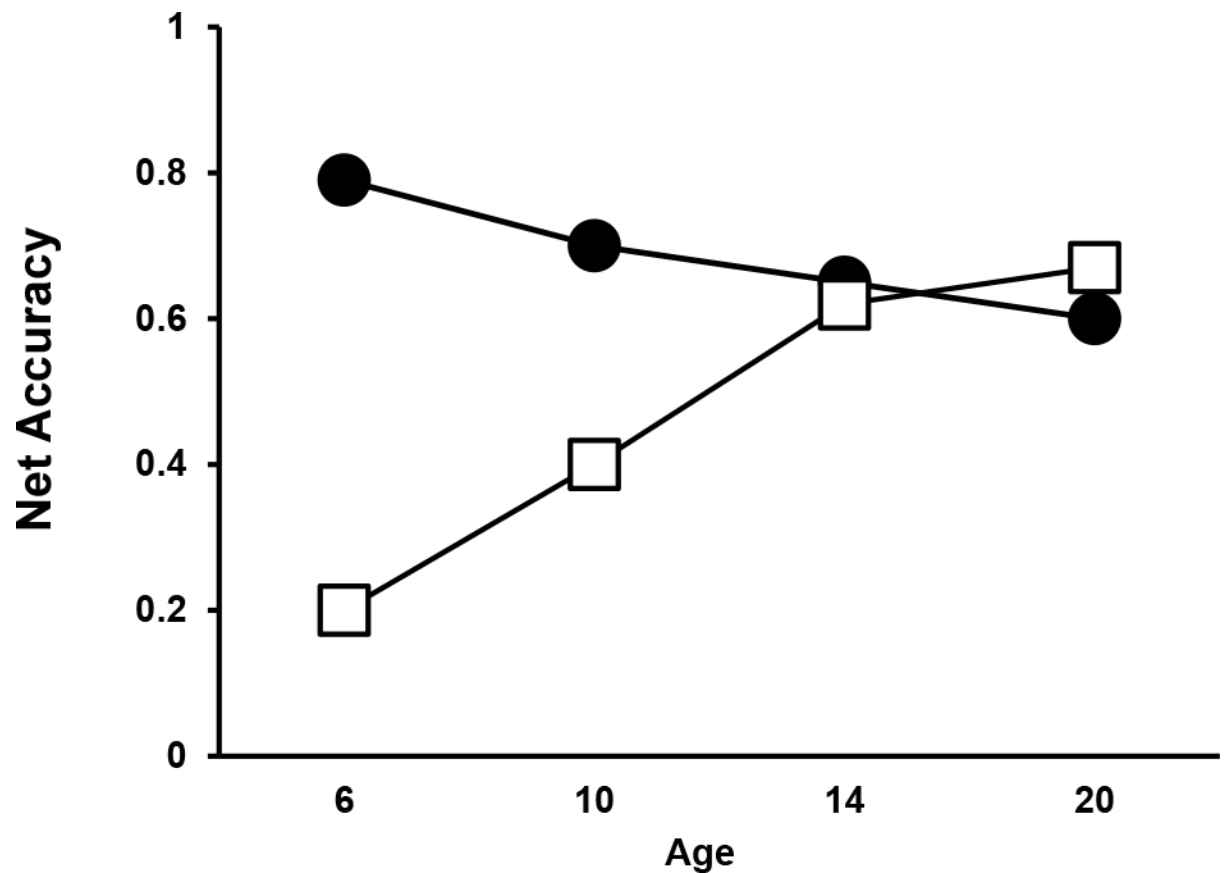


Figure 4.

Complementary developmental trends in net memory accuracy when subjects judge whether old items and new-similar items are old (black dots) versus when they judge whether the same items are new-similar (white squares). The net accuracy of old judgments is the probability of correctly recognizing list items as old divided by the sum of that probability and the probability of incorrectly recognizing new-similar items as old. The net accuracy of new-similar judgments is the probability of correctly recognizing new-similar items as new-similar divided by the sum of that probability and the probability of incorrectly recognizing list items as new-similar.

Table 1

Experimental materials and standard procedures for the Deese/Roediger/McDermott illusion

Old list words (O)	New-similar words (NS)	New-different words (ND)
<i>hot, snow, warm, winter, ...</i>	<i>cold</i> (strong), <i>frost</i> (weak)	<i>city, music</i>
<i>rest, awake, tired, dream, ...</i>	<i>sleep</i> (strong), <i>yawn</i> (weak)	<i>cold, soft</i>
<i>mad, fear, hate, temper, ...</i>	<i>anger</i> (strong), <i>mean</i> (weak)	<i>spider, thief</i>

Note. Subjects study short word lists composed of related items, such as those in the left hand column, and they respond to a recognition test on which three types of test probes are presented: old (O) list words, new-similar (NS) words, and new-different (ND) words. For each test probe, the subjects' task is to decide whether it is O.

Table 2

Mean bias-corrected correct response probabilities (SDs in parentheses) in the old (O?) and new-similar (NS?) memory test conditions of Experiments 1–4

Test items	Age level/memory test condition							
	6 years		10 years		14 years		Adult	
	O?	NS?	O?	NS?	O?	NS?	O?	NS?
Experiment 1								
Old items	.60 (.15)	.56 (.29)	.72 (.14)	.72 (.22)	.79 (.09)	.86 (.13)		
Strong new-similar items	.59 (.22)	.25 (.24)	.33 (.18)	.31 (.24)	.25 (.13)	.43 (.29)		
Weak new-similar items	.92 (.09)	.19 (.20)	.84 (.13)	.37 (.19)	.68 (.18)	.55 (.17)		
Experiment 2								
Old items	.57 (.34)	.47 (.26)	.69 (.27)	.46 (.29)	.83 (.21)	.65 (.42)	.92 (.09)	.68 (.22)
Strong new-similar items	.82 (.21)	.19 (.26)	.70 (.30)	.32 (.30)	.49 (.34)	.51 (.24)	.25 (.18)	.80 (.20)
Weak new-similar items	.93 (.18)	.14 (.24)	.76 (.25)	.24 (.34)	.67 (.34)	.29 (.36)	.57 (.32)	.61 (.28)
Experiment 3								
Old items	.55 (.25)	.49 (.22)	.70 (.22)	.53 (.32)	.85 (.11)	.66 (.26)	.83 (.14)	.67 (.15)
Strong new-similar items	.88 (.12)	.09 (.23)	.72 (.26)	.26 (.24)	.54 (.30)	.47 (.22)	.38 (.25)	.65 (.29)
Weak new-similar items	.83 (.29)	.05 (.24)	.69 (.36)	.23 (.35)	.68 (.36)	.29 (.34)	.51 (.36)	.47 (.36)
Experiment 4								
Old items	.66 (.24)	.48 (.27)	.71 (.33)	.47 (.27)	.78 (.19)	.60 (.31)	.87 (.11)	.71 (.22)
Strong new-similar items	.80 (.32)	.19 (.21)	.64 (.31)	.45 (.31)	.52 (.32)	.62 (.27)	.31 (.26)	.69 (.24)
Weak new-similar items	.85 (.23)	.05 (.15)	.75 (.31)	.20 (.27)	.67 (.34)	.44 (.33)	.52 (.34)	.50 (.32)

Note. Old items appeared on the DRM lists, whereas strong new-similar items and weak-new similar items did not appear on DRM lists. The O? and NS? testing conditions ask, respectively, whether a test item is an old list item and whether it is a new-similar item. In the O? testing condition, the correct response is acceptance for old items but rejection for strong new-similar items and weak-new similar items. In the NS? testing condition, the correct response is rejection for old items but acceptance for strong new-similar items and weak-new similar items. The correct response probabilities for old, strong new-similar, and weak new-similar test items have been corrected for yea-saying bias with the two-high-threshold method.

Table 3

Mean false alarm probabilities (SDs in parentheses) for new-different items in the old (O?) and new-similar (NS?) memory test conditions of Experiments 1–4

Experiment	Age level/memory test condition							
	6 years		10 years		14 years		Adult	
	O?	NS?	O?	NS?	O?	NS?	O?	NS?
Experiment 1	.14 (.12)	.14 (.17)	.04 (.06)	.17 (.11)	.01 (.03)	.12 (.13)		
Experiment 2	.09 (.14)	.12 (.12)	.09 (.12)	.07 (.13)	.02 (.06)	.07 (.12)	.04 (.07)	.01 (.03)
Experiment 3	.12 (.16)	.10 (.16)	.06 (.12)	.09 (.11)	.05 (.10)	.07 (.11)	.01 (.05)	.01 (.03)
Experiment 4	.09 (.14)	.12 (.12)	.11 (.12)	.08 (.13)	.02 (.06)	.18 (.24)	.03 (.07)	.01 (.02)